

SOTA AI Model Cards

A Transparency Audit & Policy Guidance

Vashishth Doshi Ziqian Zhong Saumya Goyal

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Abstract

Machine learning model cards were proposed in 2019 as a standardised mechanism for disclosing what an AI system is, how it was built, and where it fails. Seven years later, every major frontier AI laboratory publishes a model card or system card. None of them, assessed against a fine-grained rubric applied consistently across six state-of-the-art models from three leading laboratories, scores above 60 out of 100. The fields most consequential for downstream accountability — privacy risks, deception behaviour, sycophancy, fairness and bias, and hallucination rates — score below the threshold for adequate disclosure across every model in this study. This document presents the full basis for that finding: a review of the model card framework and its evolution for the large language model era; a dual methodology audit combining human reviewers and two independent LLM evaluators; and an analysis of the current legislative landscape, structured around the three distinct regulatory mechanisms through which governments are attempting to mandate model transparency without coordination. The document closes with three concrete asks addressed to advocates and policymakers working to close the gap between what model cards currently are and what accountability requires them to be.

1 Part 1: Model Cards — Origin, Purpose, and Evolving Framework

1.1 Why Documentation Matters

Machine learning models have been deployed consequentially — in hiring, in healthcare triage, in credit assessment, in criminal justice risk scoring — without standardised disclosure of what those systems do, how they were trained, or where they are known to fail. The absence of documentation is not a neutral condition. It forecloses informed decision-making by everyone downstream of the developer: the enterprise that deploys the model, the practitioner who relies on its outputs, the regulator who must determine whether it is fit for purpose, and the individual who is affected by decisions it influences. A model deployed without adequate disclosure cannot be audited, cannot be challenged, and cannot be meaningfully held accountable for its failures.

The history of AI bias incidents is, in significant part, a history of documentation failures. Demographic disparities identified only after deployment, capability boundaries discovered through use rather than disclosure, high-risk domain applications of models whose developers explicitly listed them as out of scope — these are not primarily failures of engineering. They are failures of structured communication between those who build AI systems and those who depend on them. Model cards were proposed as an answer to precisely this problem.

1.2 What a Model Card Is

Mitchell et al. proposed model cards in 2019 as a standardised reporting artifact for machine learning systems [1]. The framework is designed to communicate key information about a model in a consistent, accessible structure — not only for technical audiences, but for the full range of stakeholders who interact with or are affected by AI systems. A model card is not a marketing document and not a terms-of-service disclosure. It is a structured disclosure of what a model is, what it is for, how it performs, and where it should not be used.

The original framework organises disclosure across seven sections. **Model Details** captures the foundational provenance of the system: who developed it, when it was released, what version it represents, and what technical approach it uses. **Intended Use** specifies the application contexts the developers designed the model for, and — critically — the uses they did not design for and recommend against. **Factors** identifies the variables most likely to affect model performance across different groups or contexts. **Metrics and Quantitative Analysis** describe the evaluation approach and performance outcomes, disaggregated wherever possible across relevant subgroups. **Data** documents the training and evaluation datasets, including

known biases and provenance. **Ethical Considerations** surfaces known risks, potential harms, and the measures taken to address them. **Caveats and Recommendations** closes with explicit guidance for users about known limitations and how to work within them.

Each section serves a different stakeholder. AI developers and providers use the framework to understand what adequate disclosure requires of them. Enterprise deployers use intended use and data sections to assess what liabilities they inherit and where they need additional safeguards. Domain practitioners — clinicians, educators, civic administrators — use metrics and factors sections to understand failure modes in their specific contexts. Regulators use the overall card to determine whether documentation is sufficient for compliance and procurement assessment. And the general public, however indirectly, relies on model cards to establish that the risks to which they are exposed were identified and disclosed to someone accountable.

1.3 Antecedents and Related Frameworks

Model cards did not emerge without precedent. Several related frameworks had established the principle that transparency in consequential systems requires structured documentation, not informal disclosure. TRIPOD, introduced by Collins et al. in 2015, provided a reporting guideline for predictive models in clinical and healthcare settings, emphasising reproducibility and methodological clarity [2]. The Dataset Nutrition Label, proposed by Holland et al. in 2018, offered a structured approach to summarising dataset characteristics — making the link between data quality and model behaviour explicit [3]. Data Statements for NLP, developed by Bender and Friedman in 2018, applied documentation practice specifically to natural language processing datasets, highlighting linguistic and demographic attributes that shape model behaviour [4].

These frameworks share a common logic: that transparency must extend beyond the model itself to include the data and evaluation processes that underpin it. Model cards synthesise that logic into a single, model-level artefact. They are the point at which data documentation, evaluation documentation, and deployment guidance converge into a disclosure that any stakeholder can read and act on.

1.4 The LLM Era: What Changed

The original model card framework was designed for bounded, single-purpose machine learning systems. Large language models are neither bounded nor single-purpose. They are trained at scales that make full dataset disclosure practically impossible, deployed across an effectively unlimited range of applications, and capable of producing outputs — including harmful ones — that their developers cannot fully anticipate. The original framework does not adequately capture these characteristics. Mamirov, Azmain, and Wang identified this gap in a 2025

study that systematically evaluated documentation practices across frontier and open-source models and proposed an updated framework for the LLM era [5].

The updated framework extends the original across three areas of particular importance. **Input and Output Specifications** require detailed descriptions of what the model accepts as input and produces as output — including token counts, modality coverage, and structured versus unstructured data handling — that the original framework did not address. **Sustainability Reporting** requires disclosure of the hardware and software used for training and inference, and the environmental impact of operating at scale. This is directly relevant to procurement decisions and increasingly to regulatory compliance in jurisdictions that impose sustainability disclosure requirements. **Critical and Safety Evaluations** introduce dedicated sections addressing risks specific to large-scale models: harmful output potential, misuse scenarios, red-teaming results, and the mitigation strategies implemented in response. These sections are weighted heavily in the rubric this study applies — Safety Evaluation alone accounts for 25 percent of the total score — reflecting the degree to which safety disclosure has become the most consequential dimension of model card adequacy in the current landscape [5].

The evolution from the original to the LLM-era framework is not a technical refinement. It is a recognition that the stakes of inadequate disclosure have increased in proportion to the capability and deployment scale of the systems being disclosed.

2 Part 2: Auditing the State of the Art — Methodology and Findings

2.1 The Problem with Existing Evaluations

Existing approaches to model card evaluation share two structural limitations that this study addresses directly. The first is coarse grading. Most prior evaluations assess each section of a model card on a binary present-or-absent basis. A card that names five academic benchmarks and prints their scores without explanation is rewarded equally to one that describes the evaluation methodology, contextualises the results, and identifies failure modes. This coarseness does not merely fail to distinguish good disclosure from poor disclosure. It actively rewards what the rubric applied here calls *safety-washing*: acknowledging a topic without providing any information of substance. The Atlas study conducted by Mamirov et al. on 100 model cards from Hugging Face found that 72 percent of cards lacked any meaningful safety evaluation content — but binary grading would record many of them as having the section present [5].

The second limitation is the absence of qualitative rationale. Checklists and numeric scores identify what is missing at a headline level. They do not explain what is missing, why it matters, or what improvement would look like. A reviewer who can only record present or absent cannot surface the finding that a model card defers an entire risk section to a predecessor document, or that its refusal evaluation is conducted on a single-turn internal dataset that does not account for adaptive multi-turn approaches. Qualitative reviewing can. This study requires qualitative commentary from every reviewer, human or automated, as a first-class output alongside numeric scores.

2.2 Rubric Design

The rubric applied in this study grades every model card against eight top-level sections and 36 subsections derived from the LLM-era framework of Mamirov et al. [5]. Section-level weights are: Safety Evaluation (25%), Critical Risk (20%), Model Details (15%), Model Data (15%), Intended Use (10%), Model Inputs and Outputs (6%), Risk Mitigations (4%), and Implementation and Sustainability (5%). These weights are not adjusted for this study; they are applied as specified in the source framework.

Each subsection is scored on a five-point scale from 0 to 4. The rubric is deliberately conservative: when a card sits between two scores, the lower one is assigned. Volume of text alone does not earn credit. Table 1 presents the scale legend.

The most consequential cut in this rubric is the boundary between 1 and 2. A binary

Table 1: Five-point rubric scale with scoring anchors.

Score	Label	Meaning
0	Absent	Not addressed at all, or only named in passing with no content behind the mention.
1	Safety-washing	Topic acknowledged but disclosure is shallow. Running five academic benchmarks and printing the scores belongs here, not at 2.
2	Reasonable	Includes the what, the how, and the result with enough specificity that a practitioner could critique the methodology or partially reproduce it.
3	Good	Reasonable plus genuine interpretation: meaning of results, weaknesses, comparisons, and explicit scope. Represents current best industry practice.
4	Genuinely surprising	Goes meaningfully beyond what any frontier lab is currently doing: open evaluation artifacts, candid disclosure of unflattering findings, novel methodology, or contributed public benchmarks.

evaluation collapses scores of 1 and 2 together, recording both as present. The empirical effect of this collapse is demonstrated in Section 2.6.

2.3 Subject Pool

This study evaluates six model cards from three frontier laboratories, selected to enable within-lab comparison across a major and a minor release:

- **xAI:** Grok 4 Model Card; Grok 4.1 Model Card
- **Anthropic:** Claude Opus 4.6 System Card; Claude Sonnet 4.6 System Card
- **Google DeepMind:** Gemini 3 Pro Model Card; Gemini 3.1 Pro Model Card

The within-lab pairs allow analysis of documentation regression on minor releases, which Section 2.9 addresses. For Gemini cards, the documents link to supplementary PDFs — a Frontier Safety Framework report and a model evaluation report. Human reviewers consulted these linked documents; LLM reviewers received the same supplementary materials and were instructed to credit them only where the main card explicitly referenced them.

2.4 Evaluation Protocol

Human evaluation proceeded in two phases. In Phase 1, three of the authors divided the cards by laboratory, with each reviewer anchoring their scoring across the within-lab pair. This reduced reading burden and improved within-lab consistency. In Phase 2, a wider pool of

voluntary reviewers applied the same rubric independently. Filling out a single card carefully takes an hour or more; consequently, most respondents scored only one or two cards, and no respondent attempted all six. The final dataset contains 21 evaluations from 14 distinct reviewers across the six cards, with per-card sample sizes ranging from three to five reviewers.

This design imposes two structural caveats. First, sparse coverage: no reviewer evaluated more than two cards, so cross-lab comparisons must be inferred from aggregated reviewer pools rather than any individual reviewer’s judgment. Second, rater-bias variance: voluntary participation produced a 42-point spread in per-reviewer leniency on a 100-point scale. On a per-card basis, inter-rater spread is often larger than between-card differences. The study addresses this through within-reviewer pairwise comparison and bias-adjusted aggregate means, both described in Section 2.8.

Automated evaluation uses two independent LLM evaluators: OpenAI GPT-5.5 via the Codex CLI with reasoning effort set to medium, and Anthropic Claude Opus 4.7 via the Claude CLI with effort set to extra-high. Both engines receive the same prompt: a verbatim copy of the rubric and scoring legend, plus the model card document. Each engine produces a scores file and a comments file independently. The implementation is publicly available at <https://github.com/fjzzq2002/model-card-scoring-course-project>.

2.5 Comparison Axes

The study reports three numeric axes per card and one ranking axis. Together they are designed to isolate different sources of signal and to make the findings robust to methodological choices that any single axis could not survive alone.

Full weighted score (out of 100). Each subsection contributes $(\text{score}/4) \times \text{weight}\%$. This is the primary metric. It captures both whether a topic is present and how deeply it is treated — the variance that binary grading discards.

Binary score (out of 100). Each subsection contributes its full weight if the score is ≥ 1 , and zero otherwise. This collapses the rubric to present-or-absent and reproduces the kind of grading that Mamirov et al. critique [5]. Reporting it alongside the full score provides a direct test of how much information fine-grained grading adds: if the two axes rank cards identically, the extra granularity is not distinguishing the cards. Section 2.6 shows that it is.

Within-reviewer pairwise preference. When a reviewer scored two cards — always a within-lab pair — the sign of their score difference is compared to the sign of each LLM’s score difference on the same pair. This isolates relative judgment from rater-lenency confounds. A reviewer whose absolute scores are 20 points above average still produces a reliable signal

about which of two cards they preferred.

Bias-adjusted means. A model of the form $\text{score}_{ij} \approx \mu_j + b_i$ is fitted — card quality plus reviewer bias — by alternating least squares with the constraint that reviewer biases sum to zero. Reviewers who scored only one card still contribute to identifying card means once their bias is jointly estimated. The resulting μ_j values are reported as the bias-adjusted human aggregate in Section 2.8.

2.6 Headline Results

Table 2 reports the weighted full score and the binary score for each card under all three reviewer sources — Human (H), GPT-5.5 (G), and Claude Opus 4.7 (C). Human values are reviewer means with sample size in parentheses. LLM values are single deterministic runs at the specified effort settings.

Table 2: Per-card scores (out of 100), sorted by human full mean. Bold indicates the primary human mean column.

Card	n	Full H	Full G	Full C	Binary H	Binary G	Binary C
Opus 4.6 System Card	4	56.3	54.6	44.8	86.1	95.0	90.0
Sonnet 4.6 System Card	3	52.3	45.6	41.1	81.8	92.0	88.0
Gemini 3.1 Pro Model Card	5	50.5	42.9	43.5	85.4	93.0	93.0
Gemini 3 Pro Model Card	3	47.3	41.5	42.5	92.0	95.0	95.0
Grok 4 Model Card	3	41.1	36.3	28.0	80.8	84.5	77.0
Grok 4.1 Model Card	3	39.3	33.9	27.5	72.8	81.0	72.5

Two findings stand out. First, rankings agree broadly across reviewer sources. Spearman’s correlation between the human mean and GPT-5.5 is $\rho = +1.00$, a perfect rank match across all six cards. Correlation between the human mean and Claude Opus 4.7 is $\rho = +0.83$. The only rank disagreement is that Claude Opus 4.7 places the Sonnet 4.6 card below the Gemini cards. Second, automated evaluators are systematically stricter than human reviewers. The average raw subsection score on the 0-to-4 scale is 1.93 for humans, 1.46 for GPT-5.5, and 1.30 for Claude Opus 4.7. This gap is consistent across every section, not concentrated in any one area, and reflects reviewer leniency rather than section-specific disagreement.

The binary column exposes the most important structural finding of the study. Binary scores cluster in a 23-point band (72.5 to 95.0), while full scores span a 29-point band (27.5 to 56.3).

Gemini 3 Pro scores 92–95 on binary across all reviewers — virtually identical to Opus 4.6 — but only 41.5–47.3 on the full rubric. The binary axis is answering the question: did the laboratory mention every topic at least once? Frontier laboratories almost always have. The variance that matters is in how much depth each topic receives, and the binary axis cannot see it.

2.7 What the Comments Reveal

The qualitative comment tables collected from human reviewers — and independently from each LLM — surface findings that numeric scores alone cannot. Three categories of finding emerge.

Stylistic divides across laboratories. Anthropic cards concentrate their disclosure quality in safety evaluation. The Opus 4.6 card received nine separate score-of-4 ratings on safety-related subsections from independent reviewers. Gemini cards excel in inputs and outputs coverage and in intended use specification, providing detail that the Anthropic and xAI cards largely omit. Grok cards offer comparatively little outside critical risk evaluations. One reviewer’s comment on the Grok 4.1 safety section summarises the finding: the refusal evaluation uses a single-turn internal dataset that does not account for adaptive or multi-turn approaches; beyond that, the card relies on standard academic benchmarks; and the AgentHarm score, combined with increases in deception and sycophancy metrics, is described without red-teaming evidence.

Robust score-of-4 callouts. Across all 21 human evaluations, 80 individual score-of-4 ratings were recorded. Filtering for cross-reviewer corroboration — defined as a score of 4 from two or more independent reviewers, excluding corroboration from the two most lenient reviewers in the dataset — produces a much cleaner signal. Table 3 lists the eight subsections that meet this standard.

Candid disclosure as an exception worth naming. Several reviewers flagged unusual candour about uncomfortable findings as the most notable content of a card. The Gemini 3.1 Pro card openly discloses a cyber alert threshold breach — described by one reviewer as rare and commendable. The Opus 4.6 card states explicitly that it is not possible to evaluate with certainty whether the model is safe against CBRN misuse. A healthcare-domain reviewer noted that the Opus 4.6 card’s disclosure of strategic deception behaviour was the most useful piece of information in the card: if the model might misrepresent data after a certain point, that is essential knowledge for anyone relying on it in a clinical context. These disclosures represent what a score of 4 looks like in practice: not the absence of concerning findings, but the presence of honest ones.

Table 3: Robust score-of-4 callouts (≥ 2 independent reviewers). Gemini 3 Pro received zero score-of-4 ratings from any reviewer.

Card	Section	Subsection
Opus 4.6	Critical Risk	CBRN
Opus 4.6	Critical Risk	Cyber Risk
Opus 4.6	Safety Evaluation	Disallowed Content Handling
Opus 4.6	Safety Evaluation	Adversarial Robustness
Opus 4.6	Safety Evaluation	Red Teaming Results
Opus 4.6	Inputs & Outputs	Token Count
Sonnet 4.6	Safety Evaluation	Fairness & Bias Evaluations
Sonnet 4.6	Model Details	Organisation developing the model

2.8 Inter-Reviewer Variance and Bias-Adjusted Means

Reviewer leniency varies substantially. Fitting individual reviewer biases against model means produces a 42-point spread, from -18.4 to $+23.6$ on a 100-point scale. On any given card, two reviewers commonly differ by more than 30 points. The inter-reviewer spread on the same card is often larger than the between-card differences the study is trying to measure. This makes any single reviewer’s absolute score unreliable as a standalone readout.

Two analyses address this. Within-reviewer pairwise preference compares each reviewer’s preferred card within a within-lab pair to each LLM’s preference on the same pair. GPT-5.5 and Claude Opus 4.7 each agree with the human preference on five of seven pairs. The two disagreements both involve the Gemini pair: all three reviewers who scored both Gemini cards prefer Gemini 3 Pro, while both LLMs place Gemini 3.1 Pro slightly higher — most likely because the auxiliary PDFs admitted for the Gemini 3.1 Pro run are largely Gemini 3 Pro’s documents, so the LLM credits 3.1 Pro for content the 3.1 Pro card does not itself surface.

Bias-adjusted means, produced by fitting a model that separates card quality from reviewer leniency, change the ranking in one notable way: Gemini 3 Pro moves to second place overall. Its three reviewers all turn out to be on the harsher end of the leniency distribution, so the raw mean understates the card. Table 4 reports both raw and adjusted means.

Three distinguishable mechanisms drive reviewer bias. **Effort:** reviewers who write minimal qualitative commentary tend to give high scores because noticing what a card does not say requires more careful reading than crediting what it does. **Calibration:** some reviewers apply the scale bimodally, treating it as binary present-or-absent rather than using the full five-point range, despite writing substantive qualitative feedback. **Persona fit:** a healthcare-domain

Table 4: Bias-adjusted human means versus raw means and LLM scores.

Card	Raw mean	Adjusted mean	GPT-5.5	Claude 4.7
Opus 4.6	56.3	57.1	54.6	44.8
Gemini 3 Pro	47.3	56.3	41.5	42.5
Sonnet 4.6	52.3	55.0	45.6	41.1
Gemini 3.1 Pro	50.5	47.6	42.9	43.5
Grok 4.1	39.3	42.1	33.9	27.5
Grok 4	41.1	41.5	36.3	28.0

reviewer and a model-developer reviewer scoring the same card will emphasise different sections and find different amounts of value in the same disclosure, producing score distributions that reflect their information needs as much as the card itself. Any evaluation programme with fewer than three reviewers per card, or reviewers drawn from a single persona, risks producing scores dominated by reviewer-side variance.

2.9 Regression on Minor Releases

A consistent pattern emerges across all three within-lab pairs: minor releases score lower than the major release they follow. The Grok pair shows this most clearly — every evaluator, human and automated, scores Grok 4.1 below Grok 4, with no auxiliary documentation to complicate the comparison. The Opus/Sonnet pair shows the largest automated regression, with GPT-5.5 scoring Sonnet 4.6 nine points below Opus 4.6. The Gemini pair appears to reverse the trend under raw human means, but bias adjustment removes this reversal: Gemini 3.1 Pro falls 8.8 points below Gemini 3 Pro once reviewer leniency is accounted for.

Part of this regression is structural deferral. Sonnet 4.6 explicitly redirects to the Opus 4.6 card for many details. Gemini 3.1 Pro defers entire sections to Gemini 3 Pro. One reviewer’s comment captures the consequence: a reader of the Gemini 3.1 Pro card alone infers nothing about how risks are managed, because the entire section is deferred. Another, reviewing Sonnet 4.6, notes that when a card defers to its predecessor, even the predecessor’s qualifications get lost in the chain — Sonnet 4.6 mentions it may be safer than Opus 4.6 on CBRN, while Opus 4.6 states that CBRN safety cannot be confirmed with certainty.

But not all regression reduces to deferral. Grok 4.1 does not heavily defer to Grok 4, yet scores lower across every evaluator. Minor releases appear to receive less documentation attention than headline releases, regardless of whether they introduce novel capabilities or safety-relevant changes. Minor releases often do introduce such changes. They deserve full and self-contained disclosure, not abbreviated footnotes.

2.10 Robustness and Weaknesses

Robustness

Three independent checks support reading the results above as real properties of the cards rather than artefacts of the study design.

Two LLM engines, run independently, agree on the headline ranking. GPT-5.5 reproduces the human rank order exactly, with Spearman’s $\rho = +1.00$. Claude Opus 4.7 reproduces it with one swap, at $\rho = +0.83$. The two engines disagree on absolute scores — Claude is consistently approximately seven points harsher — but agree on the differences between cards. Cross-engine consistency makes engine-specific bias an unlikely explanation for the ordering.

Rankings are robust to plausible reweighting. As a sensitivity check, all scores were recomputed under an alternative weighting that reduces Model Details from 15% to 8% — down-weighting subsections trivially findable on laboratory marketing pages, including model overview, organisation, version, release date, and paper links — and redistributes the freed weight toward Risk Mitigation (+2 points), Implementation and Sustainability (+2 points), Intended Use (+1 point), Critical Risk (+1 point), and Inputs and Outputs (+1 point). Under this alternative weighting the Spearman ranking is identical and per-card totals shift by less than 1.5 points in every case. The headline ordering does not depend on the specific weights chosen.

Within-reviewer pairwise preferences agree with both engines on 5 of 7 pairs, even though absolute scores differ widely between reviewers. The cross-card direction signal is preserved even when reviewer leniency is not.

Weaknesses

The genuine limitations of this study, stated explicitly so that future work can address them, are four in number.

Small per-card sample. With $n = 3$ –5 reviewers per card, standard errors on absolute means are loose. The bias-adjusted aggregate is consistent in expectation but reviewer-bias estimates for single-evaluation reviewers are particularly uncertain.

Limited LLM panel. Only two engines are used, at fixed effort settings — GPT-5.5 at medium reasoning effort and Claude Opus 4.7 at extra-high. Generalisation beyond these specific configurations is untested. Running additional engines or sweeping effort levels would tighten the LLM-side variance estimates.

Voluntary, self-selected reviewer pool. Phase 2 respondents were unpaid and chose which cards to score. The persona distribution in the final sample is therefore not representative of any specific stakeholder group. Researchers and model developers are over-represented relative to general end-users and policymakers, which may affect which failure modes the qualitative comments surface.

Auxiliary document attribution is contested. Human reviewers and LLM evaluators disagree on whether a card should be credited for content it merely links to rather than contains — the Gemini case described in Section 2.8. The decision to admit auxiliary PDFs as canonical for the Gemini cards reflects the practice of human reviewers who consult them in context, but this remains a judgment call that a future study could resolve through explicit protocol design.

3 Part 3: The Policy Landscape and What It Requires

3.1 A Sector Precedent: What SR 11-7 Demonstrated

In 2011, the Federal Reserve issued **SR 11-7**, supervisory guidance that made model documentation mandatory for every bank operating under US federal oversight [6]. The guidance was a direct response to the 2008 financial crisis, in which mathematical models — opaque, undisclosed, and insufficiently validated — made consequential decisions across interconnected institutions until the consequences cascaded across the entire system. SR 11-7 solved this through three mechanisms: *information documentation* (a model inventory record requiring training data sources, known limitations, and defined approved uses), an *accountability structure* (named model owners and independent validation by a separate Model Risk Group), and *update triggers* (annual review requirements and defined criteria for material change).

The parallel to large language models is structural, not analogical. Foundation models are mathematical black boxes making consequential decisions across every sector simultaneously. Eighty-eight percent of organisations now use AI in at least one business function. A single undisclosed model deployed across healthcare, finance, education, and civic administration carries the potential for cross-domain contagion that SR 11-7 was designed to prevent in a single sector. Model cards address information documentation — the first pillar of the SR 11-7 structure. The other two pillars — independent validation and mandatory update triggers — do not yet exist as requirements for AI systems in any jurisdiction. The policy asks in Section 3.4 are, in part, the beginning of building them.

3.2 The Legislative Ladder: Three Tiers, Three Failures

Model card transparency is not absent from law. It is present three times over, in three structurally distinct forms, each moving toward the same destination without coordination. Governments and regulators have independently arrived at the same underlying requirement — that AI developers disclose what their models are, how they were built, and what they can and cannot do. What they have not arrived at is a shared mechanism, a shared vocabulary, or a shared enforcement baseline.

The first tier — implicit incentives — operates through safe harbour rather than obligation. The NIST AI Risk Management Framework, adopted January 2023, recommends model cards within its Map and Measure functions as a tool for documenting intended use, known limitations, and performance characteristics [7]. It carries no enforcement mechanism. Texas’s Responsible AI Governance Act (TRAIGA, 2025) translates this into a legal structure by designating NIST RMF compliance as an affirmative defence against

liability — meaning developers who produce NIST-aligned documentation, including model cards, are partially shielded from litigation [8]. The incentive is real, but its activation depends entirely on whether a developer calculates that litigation risk exceeds compliance cost. For large, well-resourced frontier model developers, that calculation has historically favoured non-disclosure: litigation exposure under TRAIGA remains untested, enforcement is at zero, and the reputational cost of absent documentation has not yet materialised as a financial one. Voluntary frameworks protect willing adopters. They do not move unwilling ones.

The second tier — content mandates — is where transparency becomes enforceable. A growing cluster of US state statutes and international instruments require developers and deployers to publish specific, enumerable disclosures about their AI systems — without naming model cards as the vehicle. Colorado’s AI Act (SB 24-205, effective June 2026) requires impact assessments documenting algorithmic discrimination risk for any high-risk AI system touching employment, housing, credit, or education [9]. New York City’s Local Law 144 (in force since July 2023) requires independent bias audits with publicly disclosed results disaggregated by race, ethnicity, and sex for any automated employment decision tool [10]. Illinois’s Human Rights Act amendment (HB 3773, effective January 2026) creates civil rights liability for discriminatory AI employment decisions, making the absence of bias documentation legally material in litigation [11]. At the international level, the EU AI Act’s high-risk provisions under Annex III impose analogous documentation obligations across credit, education, employment, and critical infrastructure [12]. Laid side by side, these instruments require disclosure of training data provenance, performance disaggregation, known limitations, and intended use — the four core sections of a standard model card [1]. A developer operating across all four jurisdictions simultaneously is meeting non-identical versions of the same underlying requirement, with no shared template and no mechanism for one jurisdiction’s compliance to satisfy another’s. The content mandate tier generates more documentation than any other, and less legible transparency than any other.

The third tier — explicit mandates — is the only one where model card-equivalent documentation is required by name or direct structural specification, with minimum content defined and enforceable consequence attached. Three instruments currently occupy this tier. OMB Memorandum M-26-04 (December 2025) designates Model, System, or Data Cards as minimum transparency deliverables in all federal large language model procurement contracts, making their absence grounds for contract termination [13]. The EU’s General-Purpose AI Code of Practice (August 2025), issued under Article 56 of the EU AI Act, adopts the standard model card template as the reference compliance format for all GPAI

providers, with adherence creating a legal presumption of AI Act conformity [14]. California’s Transparency in Frontier AI Act (SB 53, effective September 2025) requires frontier developers training models above 10^{26} floating point operations to publish structured safety frameworks covering catastrophic risk prevention, incident reporting, and capability limitations [15]. SB 53’s threshold mirrors the EU AI Act’s systemic risk definition precisely, and its disclosure obligations map directly to the safety and limitations sections of a standard model card.

The coverage gap, however, is significant. OMB M-26-04 reaches only vendors contracting directly with US federal agencies [13]. The GPAI Code of Practice reaches only providers above defined thresholds accessing the EU market. SB 53 applies only to the narrowest band of frontier developers by compute threshold, and its enforcement mechanisms remain untested. The six models audited in Part 2 sit at the edge of all three perimeters simultaneously. The most visible AI systems in deployment face the most precisely written, and the most sparsely applied, documentation requirements currently in force.

Taken together, the three tiers describe a regulatory architecture that is simultaneously over-built and under-connected. Each tier addresses the enforcement failure of the tier below it — content mandates make implicit incentives binding, explicit mandates make content requirements legible — but introduces a new structural failure of its own. The implicit tier produces no documentation. The content mandate tier produces documentation that cannot be compared across jurisdictions. The explicit mandate tier produces comparable, structured documentation that reaches only a narrow slice of the models actually in deployment. The developers most capable of absorbing compliance costs face the weakest aggregate documentation pressure. What the ladder reveals is not a progression toward transparency but a distribution of regulatory burden that is inversely related to the scale of public consequence.

3.3 The Minimum Threshold: What Adequate Looks Like

The audit findings in Part 2 establish not only how current model cards score, but what adequate disclosure would require. The rubric applied in this study defines a score of 2 as the floor for reasonable disclosure — the point at which a practitioner could assess the methodology and partially reproduce it, rather than simply noting that the topic has been mentioned. Four categories carry the majority of the scoring weight and define the minimum threshold for a model card to serve its accountability function.

Safety Evaluation (25% of total score) requires, at a minimum, a described evaluation approach, results reported against a named comparison baseline, and quantification of refusal behaviour and failure modes. Vague benchmark citations with no methodology are score 1 —

safety-washing — not score 2. The reason this is the floor rather than an aspiration is that a practitioner needs to know how a model was tested, not merely that it was tested.

Critical Risk (20% of total score) requires domain-specific assessment for each risk area — CBRN, cyber, manipulation, child safety, and privacy — with threshold language and a named evaluation framework cited for each. A score of 0 on any single critical risk domain is treated as disqualifying in this rubric. It marks the absence of accountability, not merely the absence of information.

Model Data (15% of total score) requires that data types and known biases be named with processing steps described, that the knowledge cutoff be explicitly stated, and that synthetic and user-generated data sources and licensing be disclosed. Downstream deployers cannot assess fitness for purpose without knowing what the model was trained on.

Intended Use (10% of total score) requires that primary use cases be named specifically, that out-of-scope uses be explicitly listed rather than implied, that intended users be identified, and that acceptable-use terms be referenced. Without explicit out-of-scope disclosure, enterprise deployers cannot determine what liability they are assuming.

This audit, combined with the findings of the Mamirov et al. Atlas study on 100 Hugging Face model cards [5], identifies five subsections that score below 2 across every model in both studies: Privacy Risks, Sycophancy, Deception Behaviours, Fairness and Bias, and Hallucination rates. These are not edge cases. They look like structural omissions — categories where the industry has collectively decided that the disclosure cost exceeds the compliance benefit, in the absence of any mechanism that would change that calculation.

3.4 Three Asks for Coordinated Action

The legislative ladder and the audit findings converge on three concrete policy needs. None requires new legislation from scratch. All three require coordination that does not currently exist.

First: explicit naming. Content mandate jurisdictions require the substance of a model card without requiring the artefact. The result is that developers produce fragmented, jurisdiction-specific disclosures that satisfy the letter of each statute and the spirit of none. OMB M-26-04 demonstrates that naming the artefact directly — designating Model, System, or Data Cards as a defined deliverable with specified minimum content — resolves this ambiguity at a stroke [13]. Extending that naming convention into state-level content mandate statutes, by amendment or by regulatory guidance, would immediately convert existing disclosure obligations into a coherent documentation regime. A developer that names

a single well-constructed model card as its compliance artefact would satisfy Colorado, Illinois, New York City, and EU high-risk obligations simultaneously.

Second: a standardised template built into legislation. The GPAI Code of Practice is the only quasi-legislative instrument that moves beyond naming to structure — it reproduces a defined set of fields, derived from the original model card specification, that providers must populate [14]. That approach should inform binding legislation. A template adopted by reference across content-mandate jurisdictions would collapse the fragmentation problem that tier two currently creates. The Mitchell et al. framework [1], extended by Mamirov et al. for the LLM era [5], provides a ready basis for such a template.

Third: minimum qualitative depth as a concurrent best-practice standard. Explicit naming and template standardisation address form. They do not address substance. The audit conducted in Part 2 finds that even where model cards exist and cover the required sections, they routinely underspecify training data provenance, disaggregated performance evaluation, and known limitations — the fields most consequential for downstream accountability. Legislative instruments currently specify that these fields must be present. None specifies what adequate population of those fields looks like. A minimum qualitative depth standard, developed in coordination with developers, civil society, and standards bodies, would give content to the form that legislation is beginning to require, and it would establish a basis on which the structural omissions identified in this section could be challenged as inadequate rather than merely noted as absent. Americans for Responsible Innovation is positioned to advance all three asks.

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